

HeatGeo: Heat-Diffusion Manifold Distillation for Compact Text Embedding Models

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Abstract

Large text embedding models are central to semantic search, clustering, and retrieval-augmented generation (RAG), where quality depends on the geometry induced by the teacher embedding space. Existing embedding distillation methods provide useful supervision through teacher vectors, token or layer representations, pairwise preferences, and mini-batch relations. These signals, however, do not explicitly describe the intrinsic manifold structure encoded by the teacher’s neighborhoods: how local semantic regions connect, how similarity propagates through nearby examples, and which higher-order geometry should remain stable under compression. We propose **HeatGeo**, a heat-diffusion manifold distillation framework for compact text embedding models. HeatGeo caches teacher embeddings, constructs a mutual k -nearest-neighbor graph as a discrete semantic manifold, and derives multi-scale diffusion distributions that describe how semantic mass flows across this manifold. The student matches these diffusion distributions over sampled candidate neighborhoods, with a single pointwise anchor to cached teacher embeddings as the only auxiliary term. We show that a separate spectral objective is unnecessary: because the heat kernel is a spectral low-pass filter, matching diffusion at several scales already transfers the low-frequency topology that eigenvector regression would supply, and the resulting two-term objective needs no global eigendecomposition. HeatGeo therefore turns the teacher’s final embeddings into geometry-aware supervision without requiring teacher hidden states, online teacher forward passes, or changes to the student architecture. We evaluate on STS, text classification, and pair classification to study whether manifold diffusion targets improve compact embedding distillation over pointwise, layer-level, and mini-batch relational baselines.

Introduction

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. Their usefulness comes not only from individual vectors, but from the geometric structure those vectors induce: neighborhoods, semantic regions, and smooth paths between related examples. Strong recent embedding models, including LLM-based encoders, often learn rich semantic geometry but are costly to run at scale. Encoding large private collections, refreshing indexes as documents change, or serving retrieval on constrained hard-

ware therefore motivates compact student encoders that inherit the geometry of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton, Vinyals, and Dean 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, teacher-anchored layer alignment, and relational losses that match pairwise similarities within a mini-batch. These objectives provide valuable supervision at different granularities: individual examples, internal layers, sampled pairs, or local batch structure.

The geometric view suggests a complementary form of supervision. A teacher embedding space can be treated as a sampled semantic manifold: nearby examples define local tangent neighborhoods, multi-hop paths reveal semantic continuity, and low-frequency graph structure captures broader organization. Pointwise losses encourage coordinate agreement, layer-alignment methods improve internal transfer under capacity mismatch, and mini-batch relational losses preserve sampled similarity patterns. HeatGeo asks whether the student can also learn the teacher’s intrinsic manifold geometry through diffusion targets derived from the teacher embedding space.

This motivates the central question of this work: can a compact student inherit the teacher’s semantic manifold by matching heat-style diffusion on a teacher-induced neighborhood graph? The intuition is simple. If the teacher embeddings define a discrete manifold over training examples, then random walks on this manifold reveal how semantic mass flows from each anchor to nearby examples, broader semantic regions, and higher-order communities. Matching this diffusion process gives the student geometry-aware supervision that is not tied to teacher internals or a particular student architecture.

We propose **HeatGeo**, a heat-diffusion manifold distillation method for compact text embedding models. HeatGeo first caches teacher embeddings and constructs a mutual k -nearest-neighbor graph using teacher cosine similarity. This graph acts as a discrete approximation to the teacher’s se-

semantic manifold. From it, HeatGeo defines a transition matrix and precomputes multi-scale diffusion targets corresponding to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives. The student then learns a neighborhood distribution over the same candidates and minimizes a weighted KL divergence to the teacher diffusion targets.

HeatGeo adds exactly one auxiliary signal: a lightweight pointwise anchor that aligns projected student embeddings with cached teacher embeddings, acting as a boundary condition rather than the main training signal. We argue and verify empirically that a separate spectral objective, a natural candidate for transferring global topology, is redundant here. The heat kernel is a spectral low-pass filter, so matching diffusion at increasing scales already matches the graph spectrum with increasing smoothness; a truncated eigenvector-regression term distills the same structure a second time, while adding a global eigendecomposition to the precomputation.

Two design choices turn out to determine whether this objective transfers geometry at all. The graph temperature must be small enough that the diffusion targets stay far from uniform on their support, otherwise the loss degenerates into a binary neighbor/non-neighbor label. And the random walk must be lazy: a plain walk on a mutual k NN graph mixes within about two steps, so nominally distinct scales collapse onto the same distribution and the multi-scale objective silently becomes a single duplicated term.

Our contributions are:

- We formulate text embedding distillation as manifold geometry transfer, using the teacher embedding space as a discrete semantic manifold.
- We introduce HeatGeo, which distills multi-scale heat-diffusion distributions from a teacher-induced mutual k NN graph into a compact student encoder.
- We show that an explicit spectral anchoring term is redundant given multi-scale diffusion matching, yielding a two-term objective that requires no eigendecomposition, and we support this with an ablation that adds the term back.
- We identify graph temperature and walk laziness as the conditions under which multi-scale diffusion targets carry ranking information at all, and report the target-sharpness diagnostics that make this checkable before training.
- We provide an evaluation protocol for HeatGeo across semantic textual similarity, text classification, and pair classification benchmarks, with ablations for diffusion scale, auxiliary terms, candidate sampling, and curriculum.

Related Work

Text Embedding Models and Evaluation. Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, semantic textual similarity, and transfer learning. Sentence-BERT made

transformer encoders practical for sentence-level similarity search by using siamese and triplet-network training (Reimers and Gurevych 2019), while SimCSE showed that contrastive learning with dropout noise or entailment supervision can produce strong sentence embeddings (Gao, Yao, and Chen 2021). Evaluation suites such as BEIR and MTEB further shifted the field from single-task similarity scores toward broad retrieval, classification, clustering, and semantic textual similarity evaluation (Thakur et al. 2021; Muennighoff et al. 2023). Continued maintenance of MTEB reflects the need for reproducible, multi-task embedding evaluation as models and datasets evolve (Chung et al. 2025).

Recent embedding systems increasingly rely on larger foundation models, richer training mixtures, and specialized retrieval objectives. BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-knowledge distillation across dense, sparse, and multi-vector retrieval modes (Chen et al. 2024). Qwen3 Embedding scales embedding and reranking models from Qwen3 foundation models (Zhang et al. 2025), and Dewey targets long-context embedding for RAG-style retrieval (Zhang, Zou, and Zhou 2025). This line of work provides stronger teachers and broader evaluation targets, but it does not by itself specify how a compact student should inherit the intrinsic geometry of the teacher embedding space. HeatGeo treats these models as black-box embedding teachers: it requires only their final embeddings, then distills diffusion geometry induced by their neighborhoods.

Distillation for Language and Embedding Models.

Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton, Vinyals, and Dean 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al. 2024; Ko et al. 2025; Anshumann et al. 2025; Kim et al. 2025; Xu et al. 2024). Cross-tokenizer methods such as DSKD and CDM address mismatches between teacher and student vocabularies or tokenization schemes (Zhang et al. 2024b; Chen et al. 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. Gecko distills LLM-generated and LLM-relabelled retrieval data into compact text embedding models (Lee et al. 2024). Jasper and Stella use multi-stage embedding distillation and Matryoshka Representation Learning to improve both quality and dimensional flexibility (Zhang et al. 2024a). LEAF aligns compact embedding students to teacher representations and supports asymmetric retrieval in which queries and documents may be encoded by different models (Vujanic and Ruecksties 2025). Jina Embeddings v5 combines task-targeted distillation with contrastive training to produce compact multilingual embeddings robust to truncation and quantization (Akram et al. 2026). These methods establish distilla-

tion as a central recipe for embedding model development. HeatGeo is complementary to this line: rather than introducing new teacher internals or task labels, it converts cached final embeddings into manifold diffusion targets.

Local and Relational Supervision. Pointwise embedding distillation directly matches a projected student embedding to the corresponding teacher embedding. This is simple and architecture-agnostic, but it asks a compact student to reproduce the teacher’s absolute coordinates even when the student’s capacity and embedding dimension differ. Contrastive sentence-embedding distillation methods such as DistilCSE improve the training signal by transferring contrastive behavior and pairwise preferences (Gao et al. 2021; Xu et al. 2023). Relational knowledge distillation further argues that students should preserve relations among examples rather than only individual outputs (Park et al. 2019). In embedding training, this idea commonly becomes mini-batch similarity matching: teacher and student pairwise similarity matrices are compared over the examples that happen to co-occur in a stochastic batch.

These local objectives better reflect retrieval than pure coordinate regression, but their supervision is bounded by the sampled batch or pair set. They do not explicitly model multi-hop semantic paths, community structure, or the way probability mass should spread from an anchor across the teacher manifold. Token-level and layer-level alignment methods, including CKA-style representation matching, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful intermediate supervision when teacher internals or student layers are available (Kornblith et al. 2019; Alqahtani et al. 2021; Dao et al. 2026). In particular, teacher-anchored layer alignment targets internal transfer and capacity mismatch, while HeatGeo targets the geometry induced by final teacher embeddings. These objectives operate on different aspects of distillation and can be viewed as complementary.

Graph, Manifold, and Diffusion Perspectives. Graph-based manifold learning studies how neighborhood graphs reveal lower-dimensional structure in high-dimensional data. Laplacian Eigenmaps use graph Laplacian eigenvectors to preserve local neighborhoods in low-dimensional representations (Belkin and Niyogi 2003), and Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity and geometry (Coifman and Lafon 2006). Knowledge distillation on graphs also studies how structural information can be transferred between graph models or from graph teachers to simpler students (Tian et al. 2025; Zhang et al. 2022). These works show that graph structure and diffusion can encode information unavailable from isolated examples.

HeatGeo draws on this graph-manifold view but applies it to a different distillation setting. The graph is not an observed input graph, and unlike Laplacian Eigenmaps we do not use spectral coordinates as an output at all: the diffusion-map view makes the eigenvectors implicit, since matching the heat kernel across scales is matching the spectrum under a low-pass filter. Instead, the graph is induced by a strong text embedding teacher and diffusion distributions become

soft manifold-neighborhood targets for the student. This gives compact text encoders an explicit geometry-aware supervision signal derived from the teacher embedding manifold, without a global eigendecomposition.

Problem Setup

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ denote a training corpus. A teacher encoder T maps each input to a teacher embedding

$$t_i = T(x_i) \in \mathbb{R}^{d_T}, \quad (1)$$

and a compact student encoder S_θ maps the same input to

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \quad (2)$$

The goal is to train S_θ so that the student embedding space preserves the semantic geometry induced by the teacher, especially nearest-neighbor relations, local semantic regions, and higher-order manifold structure.

Standard pointwise embedding distillation minimizes a distance between each projected student embedding and its teacher embedding:

$$\mathcal{L}_{\text{point}} = \frac{1}{N} \sum_{i=1}^N (1 - \cos(W s_i, t_i)), \quad (3)$$

where W is an optional projection. Relational distillation instead matches student and teacher similarity matrices inside a mini-batch B :

$$\mathcal{L}_{\text{rel}} = \|G_B^S - G_B^T\|_F^2, \quad (4)$$

where G_B^S and G_B^T are cosine-similarity matrices over examples in B . HeatGeo lifts the relational view from mini-batch snapshots to a teacher-induced semantic manifold represented by a neighborhood graph.

HeatGeo

Teacher Semantic Graph

HeatGeo first encodes the full training corpus with the teacher and constructs a teacher semantic graph. For each example x_i , let

$$\mathcal{N}_k^T(i) = \text{TopK}_{j \neq i} \cos(t_i, t_j) \quad (5)$$

be its top- k teacher nearest neighbors. To reduce noisy one-directional edges and hubness, HeatGeo uses a mutual k NN graph:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \quad (6)$$

The weighted adjacency matrix W^T is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where τ_g controls graph sharpness. Let D^T be the diagonal degree matrix. The row-normalized transition matrix is

$$P^T = (D^T)^{-1} W^T. \quad (8)$$

Each row P_i^T defines a one-step semantic random-walk distribution from anchor x_i .

The graph temperature τ_g is not a cosmetic hyperparameter. A mutual k NN graph has degree of order k , and teacher cosine similarities among the top- k neighbors of an anchor occupy a narrow band. If τ_g is large relative to that band, every row of P^T is close to uniform over its support, and the diffusion targets below degrade into binary neighbor/non-neighbor labels that carry no ranking information. We therefore report $D_{\text{KL}}(p_{i,r}^T \| \mathcal{U}_{\text{supp}})$, the divergence of each target from the uniform distribution on its own support, as a build-time diagnostic and select τ_g so that this quantity stays well above zero.

Multi-Scale Heat Diffusion Targets

A one-step transition captures direct neighbors, but embedding spaces also contain higher-order semantic structure. HeatGeo therefore uses multiple diffusion scales of a *lazy* random walk,

$$Q^T = \frac{1}{2} (I + P^T), \quad P_r^T = (Q^T)^r, \quad r \in \mathcal{R}, \quad (9)$$

where $\mathcal{R}$ is typically $\{1, 2, 4\}$. Small r preserves sharp local neighborhoods, while larger r captures semantic regions and multi-hop communities. Laziness is what makes $\mathcal{R}$ a graded family: a plain walk on a mutual k NN graph mixes within its local component in about two steps, so $(P^T)^2$ and $(P^T)^4$ collapse onto nearly the same distribution and the additional scales contribute duplicated terms rather than new structure. The lazy walk retains mass at intermediate step counts and yields scales that remain distinguishable. In implementation, diffusion is approximated sparsely by propagating probability mass through graph neighbor lists and retaining the top candidates, with a single walk snapshotted at each scale in $\mathcal{R}$. The truncation to a sparse support and the removal of the self-transition mass both destroy normalization, so the distribution is renormalized after each; we log the residual mass falling outside the retained support so that the approximation error is visible rather than assumed small.

For each anchor x_i , HeatGeo precomputes a diffusion pool $\mathcal{P}_i$ of size P — the retained sparse support of the walk — together with a same-source hard-negative pool $\mathcal{H}_i$ drawn from the teacher’s top-ranked non-neighbors of x_i . At the start of each epoch e it then *redraws* a candidate set

$$C_i^{(e)} = C_i^+ \cup C_i^h \cup C_i^r, \quad |C_i^{(e)}| = C \ll P, \quad (10)$$

where $C_i^+ \subset \mathcal{P}_i$ contains diffusion neighbors, $C_i^h \subset \mathcal{H}_i$ contains teacher hard negatives, and C_i^r contains random negatives, each filling a fixed slot quota. Resampling matters more than it appears. A candidate set frozen for the whole run defines a fixed finite collection of comparisons; once the student orders them correctly there is no further gradient, and training saturates after roughly one epoch regardless of how much structure the graph still holds. Redrawing $C_i^{(e)}$ exposes a different section of the same pool each epoch, so the objective keeps presenting comparisons the student has not yet fit while the underlying target distribution is unchanged. Within each scale we keep its top- m neighbors deterministically and draw the remainder of its share by Gumbel top- k sampling proportional to diffusion

mass, which reaches the tail of the support without biasing the retained head.

Diffusion positives are drawn round-robin across the scales in $\mathcal{R}$, each scale sampling from its own $p_{i,r}^T$. Drawing the whole positive quota from the mixture $\sum_r \omega_r p_{i,r}^T$ is the more obvious choice and is wrong: the mixture’s support is dominated by the broadest walk, so the sharpest scale receives one or two scored candidates and its KL term stops carrying ranking information at all. The teacher diffusion target at scale r is the diffusion mass restricted and renormalized over C_i :

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \quad j \in C_i. \quad (11)$$

This target is a soft distribution over semantic relatedness, rather than a binary positive-negative label.

Student Neighborhood Distribution

Given an anchor embedding s_i and candidate embeddings $\{s_j : j \in C_i\}$, the student induces a distribution over the same candidate set. A single student distribution, however, cannot support a multi-scale objective. Suppose the student used one temperature τ_s for all scales, giving one p_i^S . Since cross-entropy is linear in its target,

$$\sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}(p_{i,r}^T \| p_i^S) = - \underbrace{\sum_r \omega_r H(p_{i,r}^T) + H(\bar{p}_i^T, p_i^S)}_{\text{constant}}, \quad (12)$$

where $H(\cdot, \cdot)$ is cross-entropy and $\bar{p}_i^T = \sum_r \omega_r p_{i,r}^T$. Up to an additive constant the objective depends on $\mathcal{R}$ only through the single mixture $\bar{p}_i^T$: any two scale sets with the same mixture produce identical gradients, and “multi-scale” reduces to target smoothing. Worse, (12) is minimized at $p_i^S = \bar{p}_i^T$, at which point the loss equals

$$\text{JS}_\omega(p_{i,1}^T, \dots, p_{i,R}^T) = H(\bar{p}_i^T) - \sum_r \omega_r H(p_{i,r}^T) \geq 0, \quad (13)$$

the generalized Jensen–Shannon divergence of the scales. This is an *irreducible floor*: it is a property of the teacher graph, fixed before training begins, and no student can go below it. A tied-temperature run that plateaus is therefore ambiguous — the plateau may be an optimization failure or may simply be (13) — and the two have opposite fixes. We compute (13) offline from the artifact and report the *excess* $\mathcal{L}_{\text{diff}} - \text{JS}_\omega$ alongside the raw loss, so the distinction is observable at every step.

HeatGeo therefore assigns each scale its own student temperature τ_r , with broader scales receiving larger τ_r to match their smoother targets:

$$p_{i,r}^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_r)}{\sum_{j' \in C_i} \exp(\cos(s_i, s_{j'})/\tau_r)}, \quad j \in C_i, \quad (14)$$

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}(p_{i,r}^T \| p_{i,r}^S), \quad (15)$$

where B is a mini-batch, ω_r is the scale weight, and $\bar{\mathcal{R}} = \{0\} \cup \mathcal{R}$ includes the direct scale introduced in Section . The

scales now share the underlying similarity vector $\cos(s_i, \cdot)$ but not the distribution built from it, so (12) no longer applies and (13) becomes a lower bound rather than an attained value. The mechanism is over-determination: $|\mathcal{R}|$ softmaxes at distinct temperatures impose $|\mathcal{R}|$ independent constraints on one $|C_i|$ -vector of cosines, and satisfying them all requires reproducing the teacher’s similarity *gaps*, not merely its ranking. A single softmax pins $\cos(s_i, s_j)$ only up to a per-anchor additive constant, which is exactly the calibration that Spearman correlation on STS and any globally thresholded cosine retrieval depend on. In a controlled setting with free logits, minimizing the tied objective converges to the floor of (13) to four decimals, whereas untying τ_r reaches a value well below it.

The Direct Scale

The diffusion targets are graph mass renormalized over the scored columns, so every column outside $\mathcal{P}_i$ receives target *exactly zero*. A zero target is not an abstention. The gradient of the cross-entropy with respect to the logit of column j is $p_{i,r}^S(j) - p_{i,r}^T(j)$, so at a zero-target column it is $+p_{i,r}^S(j) > 0$, and the objective drives $\cos(s_i, s_j)$ down without bound. Under in-batch sharing roughly 97% of the columns an anchor is scored against are zero-target, and they include the hard negatives C_i^h — same-source, inside the teacher’s top-200 by cosine, excluded from the graph only because the mutual- k NN test rejected them. In a controlled probe with planted cluster structure, those hard negatives carry teacher cosine 0.283 against 0.349 for the diffusion positives, and the graph-only objective pushes 98% of them apart. The method is thus training the student to separate pairs its own teacher calls similar, which is precisely the calibration that STS Spearman and a global cosine threshold measure.

We therefore add a scale $r = 0$ whose target is the teacher’s own similarity over the full column set,

$$p_{i,0}^T(j) = \frac{\exp(\cos(t_i, t_j)/\tau_t)}{\sum_{j'} \exp(\cos(t_i, t_{j'})/\tau_t)}, \quad (16)$$

and optimize over $\bar{\mathcal{R}} = \{0\} \cup \mathcal{R}$. Unlike the diffusion targets, (16) is dense: every scored column receives its true teacher mass. In the same probe it assigns 0.670 to the positives, 0.271 to the hard negatives and 0.059 to the random ones, and the fraction of hard negatives pushed apart falls from 98% to 41%. The teacher embeddings are cached already, so the term is free.

Scale $r = 0$ also supersedes $\mathcal{L}_{\text{anchor}}$ on its own terms. Comparing cosines measured in the student’s metric against cosines measured in the teacher’s is *not* invariant to a linear map of the student space — an anisotropic rescaling of s changes $\cos(s_i, s_j)$ and therefore changes (16)’s loss — so this is the term that actually pins the absolute geometry, which Section shows the projected anchor cannot. Note also that τ_t cannot simply inherit τ_g : the graph temperature acts on a mutual- k NN neighborhood where cosines occupy a narrow band, whereas (16) ranges over the whole column set, where the same value would place nearly all mass on the single nearest column.

Finally, we score every anchor in B against the *union* of all candidates in the batch rather than against its own C_i alone. The candidate texts are already encoded, so this is free in forward cost, multiplies the negatives per anchor by roughly $|B|$, and — because anchors now share columns — couples their logit scales so that similarity levels become comparable across the batch. The teacher target is scattered onto the union with zero mass on the columns an anchor did not draw, and the anchor’s own column is masked.

Why No Separate Spectral Term

A natural addition would be a spectral term that regresses student embeddings onto the leading eigenvectors of the graph Laplacian $L^T = I - (D^T)^{-1/2} W^T (D^T)^{-1/2}$, on the grounds that diffusion matching is local while eigenvectors describe global topology. The usual argument for omitting it appeals to the spectral expansion of the heat kernel,

$$e^{-tL^T} = \sum_{\ell} e^{-t\lambda_{\ell}} u_{\ell} u_{\ell}^{\top}, \quad (17)$$

concluding that matching diffusion at several scales already matches the graph spectrum under a low-pass filter of bandwidth t , so an eigenvector term distills the same structure twice.

That argument does not survive Section’s own premises if the temperatures are tied. By (12), a tied objective constrains the student only through the mixture $\bar{p}^T$, i.e. through the *single* fixed filter $g(Q^T) = \sum_r \omega_r (Q^T)^r$ applied once. Observing $u_{\ell}^{\top} g(Q^T) u_{\ell} = g(1 - \lambda_{\ell})$ for one known g does not identify $\{\lambda_{\ell}\}$: g is not injective on $[-1, 1]$ for the weights we use, and in any case one scalar functional per eigenvalue recovered from row-normalized, candidate-restricted distributions is far weaker than eigenvector regression. Under a tied temperature the “already covered” claim is therefore unsupported, and the honest statement would be that the two terms are complementary.

With per-scale temperatures the argument is restored, in a weaker and more defensible form. Each scale now imposes its own constraint, so the student must match $|\mathcal{R}|$ distinct filters $(Q^T)^r$ of the same operator rather than one mixture — a filter bank, and it is the bank, not any single member, that pins down where the spectral mass sits. We still make no claim of exact spectral identifiability from $|\mathcal{R}| = 3$ scales on truncated supports; we claim only that a truncated eigenvector-regression term adds little on top of the bank while reintroducing the pipeline’s only global eigendecomposition. Section reports the ablation that adds the term back, run against the per-scale-temperature model so that the comparison is against the objective we actually advocate.

Training Objective

A relational objective constrains the student only up to a transformation of its space, which invites a pointwise anchor of the form

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|B|} \sum_{i \in B} (1 - \cos(W_a s_i, t_i)), \quad (18)$$

where W_a is a freely learned map to the teacher dimension. We report this term but disable it by default ($\lambda_{\text{anchor}} = 0$),

because as written it cannot do the job it is introduced for. For any invertible A , replacing $s_i \mapsto As_i$ and $W_a \mapsto W_a A^{-1}$ leaves (18) exactly invariant, so the term is blind to precisely the degrees of freedom — rotation, anisotropic scaling, shear — that the relational loss also leaves free. Its optimum is the best linear predictor of t_i from s_i , a quantity that a single linear layer can fit early and then hold nearly constant; empirically it settles within one epoch and contributes no further gradient of consequence. Supplying genuine absolute calibration requires either freezing W_a (e.g. to a random projection, breaking the gauge) or, as we do here, obtaining it from the per-scale temperatures, which constrain cosine gaps directly in the student’s own metric. We keep (18) only as an ablation row.

The final objective is therefore two terms with a single balancing weight:

$$\mathcal{L}_{\text{HeatGeo}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}}. \quad (19)$$

All scales in $\mathcal{R}$ are active from the first step. We also evaluated a curriculum that introduces scales one epoch at a time; because the scale weights are renormalized over the active set, such a curriculum monotonically shifts weight away from the sharpest local scale as training proceeds, and we found no benefit from it (Section).

The loss value alone is not interpretable. It cannot separate a student that has learned the teacher’s neighborhood ranking from one whose embeddings have collapsed toward a uniform candidate distribution, and by (13) it cannot separate a converged student from an exhausted objective. We therefore log, at every step, (i) the entropy of $p_{i,r}^S$ relative to $\log |C_i|$, (ii) the student probability mass falling on the teacher’s top-8 candidates, and (iii) the excess $\mathcal{L}_{\text{diff}} - \text{JS}_\omega$ over the floor computed offline from the teacher graph. Two build-time diagnostics accompany them: $D_{\text{KL}}(p_{i,r}^T \| \mathcal{U}_{\text{supp}})$ per scale, which detects a graph temperature that has flattened the targets into binary labels, and the pairwise cross-scale divergences $D_{\text{KL}}(p_{i,r}^T \| p_{i,r'}^T)$, which detect scales that have become redundant. A near-zero JS_ω is itself a warning: it certifies that the scales carry the same target and that, under a tied temperature, the multi-scale objective would be numerically identical to a single-scale one.

We also deduplicate the corpus before building the graph. Exact duplicate texts have $\cos = 1$ by construction, so their logit is pinned at the ceiling with no gradient while they continue to absorb teacher probability mass from the candidates that could still move.

Experiments

Experimental Setup

Models. Our implementation uses a compact BERT/MiniLM-style student encoder and an LLM-based teacher embedding model. Teacher embeddings are cached once before HeatGeo training. After graph construction, the teacher model is removed from GPU memory; HeatGeo training uses only cached teacher embeddings and precomputed diffusion targets.

Algorithm 1 HeatGeo Training

Require: Corpus $\mathcal{D}$, teacher T , student S_θ , graph size k , scales $\bar{\mathcal{R}} = \{0\} \cup \mathcal{R}$, temperatures $\{\tau_r\}$, teacher temperature τ_t , pool size P , candidate size C

- 1: Deduplicate $\mathcal{D}$; cache teacher embeddings $t_i = T(x_i)$
- 2: Build a mutual k NN teacher graph W^T
- 3: Compute $P^T = (D^T)^{-1}W^T$ and lazy walk $Q^T = \frac{1}{2}(I + P^T)$
- 4: Precompute diffusion pools $\mathcal{P}_i$ ($|\mathcal{P}_i| = P$) and targets $p_{i,r}^T$ from one walk snapshot at each $r \in \mathcal{R}$; precompute same-source hard-negative pools $\mathcal{H}_i$
- 5: Compute the floor JS_ω of (13) and the build-time target diagnostics
- 6: **for** each training epoch e **do**
- 7: Redraw $C_i^{(e)}$ for every i : per-scale positives from $\mathcal{P}_i$, hard negatives from $\mathcal{H}_i$, random negatives
- 8: **for** each mini-batch B **do**
- 9: Encode anchors and candidate texts with S_θ
- 10: Form the shared column set $\bigcup_{i \in B} C_i^{(e)}$ and scatter the targets onto it
- 11: Compute the direct target (16) on the shared columns from the cached t_i
- 12: Compute $p_{i,r}^S$ at temperature τ_r for each $r \in \bar{\mathcal{R}}$
- 13: Compute $\mathcal{L}_{\text{HeatGeo}}$ and the excess over JS_ω
- 14: Update θ by minimizing $\mathcal{L}_{\text{HeatGeo}}$
- 15: **end for**
- 16: **end for**
- 17: **return** Distilled student encoder S_θ

Training Data. We train on text-pair data assembled from multiple semantic tasks. Each training row provides either a text pair or a single text field depending on the task format. HeatGeo constructs the teacher graph over the training examples, using the anchor text for candidate neighborhoods.

Evaluation Tasks and Metrics. We evaluate the distilled student on three task families. For semantic textual similarity, we report Spearman correlation on SICK, STS12, and STS-B. For text classification, we freeze the encoder, train a logistic regression classifier on top of embeddings, and report accuracy and macro-F1 on Banking77, Emotion, and Tweet. For pair classification, we threshold cosine similarity and report accuracy, macro-F1, precision, recall, and average precision on MRPC, SciTail, and WiC.

Baselines. We compare HeatGeo with the undistilled student, pointwise teacher anchoring, mini-batch relational distillation, cross-tokenizer alignment methods such as CDM and DSKD, EMO-style token/attention alignment, Stella-style multi-faceted embedding distillation, TALAS teacher-anchored layer alignment, and the full teacher embedding model when feasible. These baselines cover output-level, layer-level, tokenizer-aware, and relational distillation signals.

Implementation Details. Unless otherwise specified, HeatGeo uses graph size $k = 50$, graph temperature $\tau_g = 0.05$, lazy diffusion scales $\{1, 2, 4\}$, candidate size

64, 32 diffusion positives, 16 hard negatives, 16 random negatives, and five training epochs. The scale weights are (1.0, 0.5, 0.25), giving higher weight to sharper local diffusion, and the student temperature is $\tau_s = 0.07$. The objective has a single balancing weight, $\lambda_{\text{anchor}} = 0.05$.

Target Sharpness Diagnostics. Because the informativeness of the diffusion targets depends on the teacher and the corpus, we report $D_{\text{KL}}(p_{i,r}^T || \mathcal{U}_{\text{supp}})$ per scale and the pairwise divergence between consecutive scales at graph-construction time. These two numbers reveal, before any training, whether the target carries ranking information and whether the scales are genuinely distinct; τ_g was selected on this basis rather than by downstream search.

Main Results

Table 1 reports downstream embedding quality across the evaluation tasks. The key comparison is whether manifold diffusion supervision improves compact embedding distillation relative to output-level, layer-level, and mini-batch relational baselines while retaining the efficiency of the compact student.

Ablation Study

We isolate the contributions of HeatGeo with the ablations in Table 2. Removing diffusion reduces the method to pointwise anchoring alone, and removing the anchor leaves a purely relational objective. Two ablations justify what the objective leaves out: adding a spectral term back tests the redundancy argument of Section , and adding an InfoNCE task term tests whether contrastive supervision contributes anything beyond the diffusion targets. The remaining rows probe the design choices that make the multi-scale objective non-degenerate: the plain walk against the lazy walk, larger τ_g against the selected value, the scale curriculum against activating all scales from the start, and the role of hard and random negatives.

Discussion

HeatGeo is designed around the observation that teacher embeddings define more than isolated targets: they define a semantic manifold whose diffusion geometry can serve as a distillation signal. The method has three practical advantages. First, it uses cached teacher embeddings rather than teacher hidden states, attention maps, or online teacher forward passes, which lowers training-time memory pressure. Second, it is tokenizer-agnostic because the loss is defined over final embeddings and graph diffusion targets. Third, it does not change the student architecture, so the distilled model can be deployed as an ordinary text encoder.

The main limitation is that HeatGeo requires a graph pre-processing step before student training. Building a teacher graph is worthwhile when the same training data supports substantial distillation, but it adds overhead compared with purely mini-batch objectives. HeatGeo also depends on the quality and coverage of the teacher graph: if the teacher embeddings contain spurious neighborhoods, diffusion may propagate those errors. Future work can study approximate

nearest-neighbor construction, adaptive diffusion scales, and online graph refresh for changing data.

Conclusion

We introduced HeatGeo, a heat-diffusion manifold distillation framework for compact text embedding models. HeatGeo treats cached teacher embeddings as samples from a semantic manifold and distills multi-scale lazy diffusion over a teacher-induced mutual k NN graph, with a single pointwise anchor as its only auxiliary term. Because the heat kernel is itself a spectral low-pass filter, this two-term objective transfers low-frequency topology without a separate eigenvector-regression loss and without any global eigendecomposition. This gives the student a geometry-aware supervision signal while requiring only cached teacher embeddings and no student architecture changes. HeatGeo is therefore a practical manifold-oriented complement to output-level, layer-level, and mini-batch relational embedding distillation objectives.

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Table 1: Experimental results on nine datasets. Datasets marked with $\star$ are in-domain, while the remaining datasets are out-of-domain.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg.
	Banking77	Tweet	Emotion*	MRPC	SciTail	WiC*	SICK	STS12	STSB*	Avg-In	Avg-Out	
BGE-M3 → BERT-base												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	—	—	—	—	—	—	—	—	—	—	—	—
DSKD	—	—	—	—	—	—	—	—	—	—	—	—
Jasper and Stella	—	—	—	—	—	—	—	—	—	—	—	—
DistillCSE	—	—	—	—	—	—	—	—	—	—	—	—
EMO	—	—	—	—	—	—	—	—	—	—	—	—
TALAS	—	—	—	—	—	—	—	—	—	—	—	—
HeatGeo	—	—	—	—	—	—	—	—	—	—	—	—
Qwen-Embedding-0.6B → BERT-base												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	—	—	—	—	—	—	—	—	—	—	—	—
DSKD	—	—	—	—	—	—	—	—	—	—	—	—
Jasper and Stella	—	—	—	—	—	—	—	—	—	—	—	—
DistillCSE	—	—	—	—	—	—	—	—	—	—	—	—
EMO	—	—	—	—	—	—	—	—	—	—	—	—
TALAS	—	—	—	—	—	—	—	—	—	—	—	—
HeatGeo	—	—	—	—	—	—	—	—	—	—	—	—
Qwen-Embedding-4B → BERT-base												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	73.83	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	92.34	73.74	64.41	85.97	82.66	68.65	78.38	72.27	79.98	71.01	80.89	77.60
HeatGeo	92.62	72.27	66.42	83.57	89.15	66.43	79.69	70.43	81.79	71.55	81.29	78.04

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Table 2: Ablation template for HeatGeo. Report average score across task groups or a selected validation suite. The first block ablates the two terms of the objective, the second tests whether the terms we deliberately excluded would add anything, and the third probes the conditions under which the multi-scale targets are non-degenerate.

Variant	Average Score
HeatGeo full	–
w/o diffusion loss $\mathcal{L}_{\text{diff}}$	–
w/o pointwise anchor $\mathcal{L}_{\text{anchor}}$	–
+ spectral anchoring term	–
+ InfoNCE task term	–
plain walk instead of lazy walk	–
$\tau_g = 0.1$ instead of 0.05	–
+ diffusion-scale curriculum	–
w/o hard negatives	–
w/o random negatives	–
single scale $r = 1$	–
single scale $r = 4$	–

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